Design Project 1 Summary Report

ENSC 350 – Group 08

Serene Estacio - 301592758

Duncan Miller - 301587707

Laxmi Unnithan - 301393115

Nazim Mete Balci – 301578293

Introduction

The purpose of the Design project is to design, implement and evaluate different 64-bit adder architectures using VHDL, focusing on their function and cost. The project involves four design candidates using Baseline Ripple, Fast Ripple, Conditional Sum, implemented on Cyclone IV and Arria II FPGAs.

Design Candidate Architectures Description

The Baseline uses interconnected full adders with sequential carry propagation, serving as the reference design. Fast Ripple uses the VHDL + operator, allowing Quartus to optimize the addition circuitry. Conditional sum uses a recursive structure that calculates possible sums in parallel and selects the correct result based on the incoming carry. Carry lookahead uses generate and propagate signals to calculate carries in advance, while Carry Select divides the adder into blocks that compute results for both possible carry-in values, selecting the correct result once the carry is known.

Functional Verification

Functional Verification is performed using VHDL testbenches in ModelSim. The testbenches read pre-generated testvectors from Adder00.tvs and compare the actual Sum, Carry out, and overflow with expected results.

Summary of Cost Analysis Procedure and Results

- Evaluating and comparing resource utilization and device costs

Results

BaselineTB: takes about 10,000 ps of start-delay while the .tvs file is read. There is then a 1000ps allotted per operation representing the clock signal of the TB. During this 1000 ps there is 100ps allotted to resetting all values back to their uninitialized states.

Project Conclusions and Takeaways

To achieve the project objective, we designed and evaluated six distinct 64-bit adder architectures across different Intel FPGA devices to quantify the hardware trades-offs of high-speed arithmetic logic.

Our key findings for this is the speed-area tradeoff, where parallel carry schemes have area penaltys on regular 4-input logic element architectures and increases the area overhead. The hardware cost of increasing speed is dictated by the overall fabric of the FPGA boards, as the Cyclone IV has increased logic cell replication for more complex routing and the Arria II Adaptive Logic Modules allow adders to compress down their cost.